

1 SUMMARY

Turbulence naturally generates small-scale structures in collisionless plasmas, including magnetic reconnection. Quantifying reconnection's contribution to energy dissipation requires systematic event identification — a challenging and time-consuming task with traditional methods.

Using an unsupervised machine learning approach, we identify physically distinct structures in turbulent MMS magnetosheath observations. One cluster shows signatures consistent with magnetic reconnection and recovers most known events. The reconnection-like cluster is associated with enhanced intermittency and local energy transfer at kinetic scales.

Our approach successfully recovers known reconnection events and identifies new candidates, enabling systematic studies of reconnection's role across different turbulent environments.

2 TURBULENCE-DRIVEN MAGNETIC RECONNECTION

Turbulent plasmas exhibit chaotic motion and fluctuations in electromagnetic fields across scales, generating small-scale structures such as thin intense current sheets where **magnetic reconnection** can occur.

- In collisionless plasmas, reconnection happens when sheared magnetic field lines converge, break the frozen-in condition, and reconnect.
- Stored magnetic energy is transferred to particles, accelerating ions and/or electrons at Alfvénic speeds away from the **Ion and Electron Diffusion Region** (IDR and EDR), respectively (**Fig. 1a**).

A natural laboratory for these plasma studies is the Earth's magnetosheath (**Fig. 1b**) from which NASA's *Magnetospheric MultiScale* (MMS) mission offers multi-point, high-resolution observations.

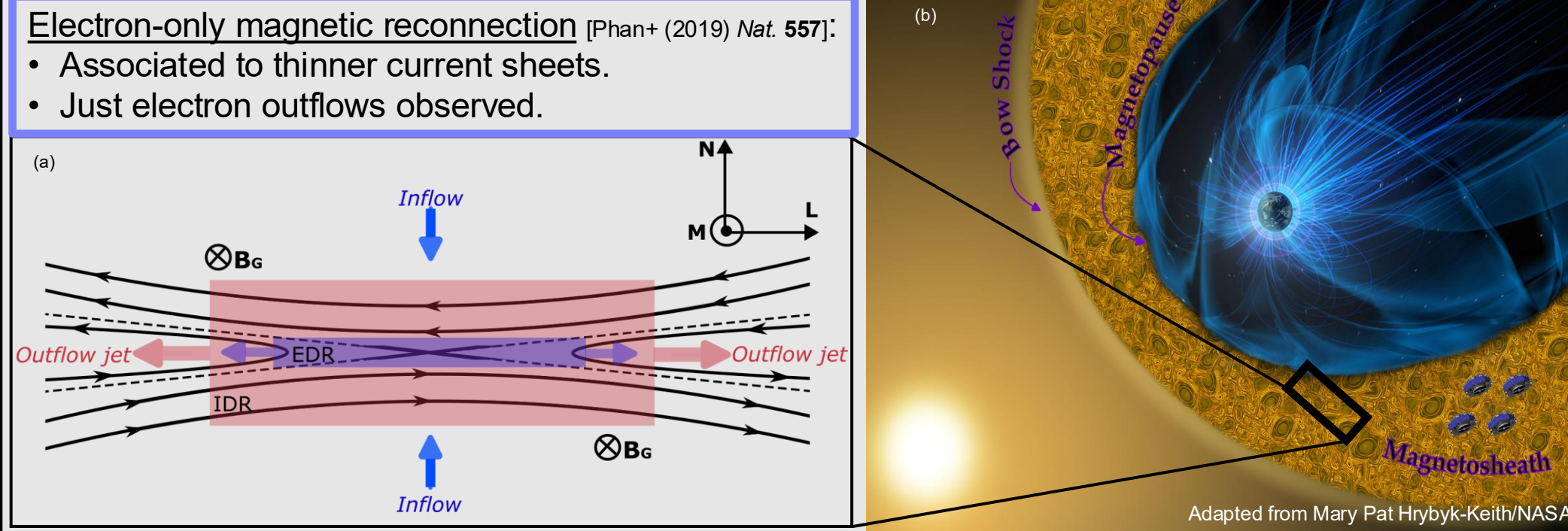


FIGURE 1. (a) Magnetic reconnection diagram. (b) Earth's turbulent magnetosheath.

Why does it matter? Reconnection and other coherent structures are thought to influence turbulence dynamics, energy dissipation, and particle heating [Stawarz+ (2022) *PhPI* 29; Shay+ (2018) *PIPh* 25]. However, the extent of reconnection's role remains uncertain → motivating systematic analysis of its occurrence and properties across turbulent environments.

Challenge: These small-scale structures generated by turbulence are localised, involve complex magnetic topologies and have random orientations → making them difficult and time-consuming to identify manually. This motivates an automated, data-driven approach.

3 AUTOMATIC IDENTIFICATION OF MAGNETIC RECONNECTION

We use an unsupervised machine-learning framework to systematically identify magnetic reconnection-related structures in turbulent MMS observations.

- The method is based on **Toeplitz Inverse Covariance Clustering (TICC)** [Hallac+ (2017) *CoRR*], which groups short windows of multivariate time series according to their correlation structure.

The workflow consists of four steps.

1. Feature Selection

- Physics-motivated features, sensitive to magnetic reconnection in turbulent observations (**Tab. 1**).
- Independent of the coordinate system.

2. Training TICC

- TICC hyperparameters were tuned using 60 reference reconnection events [Stawarz+ (2022) *PhPI* 29].
- The optimal configuration was selected by maximising the recovery of known events into a single cluster containing compact, localised structures.

3. Cluster Characterisation

TICC assigns each timeseries point to one of three clusters. Consecutive points in the same cluster define individual structures. Based on feature distributions (**Fig. 2**) and structure morphology (**Fig. 3a**), we interpret the clusters as:

- 0 (blue):** background plasma fluctuations,
- 1 (orange):** localised current enhancements frequently overlapping with known reconnection events,
- 2 (green):** broader enhanced-activity regions.

4. Reconnection Identification

- Cluster 1 (orange) recovers known reconnection events [Stawarz+ (2022) *PhPI* 29].
- Additional structures in Cluster 1 exhibit similar signatures across the TICC input features, for example, enhancements in AG_e and $\mathbf{J} \cdot \mathbf{E}'$ (**Fig 3b-3e**).
- Coordinate-dependent signatures (B_L reversal and ion/electron velocity perturbations) will further confirm whether these candidates are indeed reconnection events.

Feature name	Definition
Current density (Fig. 3a)	$ \mathbf{J} = nq v_i - v_e $
Normalised curvature of the magnetic field ($R_C = \frac{1}{ b \cdot \nabla b }$), [Rogers+ (2023) <i>JGR</i> 128]	$K_e = \log_{10} \sqrt{\frac{R_C}{\rho_e}}$
Agyrotropy of the ion and electron pressure tensor [Che+ (2018) <i>PhPI</i> 25]	$AG_{L,e} = \frac{4 \det(\mathbb{P}) - P_{\parallel}(tr(\mathbb{P}) - P_{\parallel})^2}{4 \det(\mathbb{P}) + P_{\parallel}(tr(\mathbb{P}) - P_{\parallel})^2}$
Energy conversion	$\mathbf{J} \cdot \mathbf{E}' = \mathbf{J} \cdot (\mathbf{E} + \mathbf{v}_e \times \mathbf{B})$
Non-ideal electric field	$ \mathbf{E}' = \mathbf{E} + \mathbf{v}_e \times \mathbf{B} $
Hall electric field	$ \mathbf{E}_{Hall} = \frac{ \mathbf{J} \times \mathbf{B} }{qn}$
Lorentz indicator, $v_L = c \frac{\mathbf{E} \times \mathbf{B}}{B^2}$, [Lapenta+ (2021) <i>ApJ</i> 911]	$\mathcal{L} = \log_{10} \left(\frac{v_L}{c} \right)$

Table 1: Selected features used as input for TICC, computed from MMS observations of the Earth's magnetosheath.

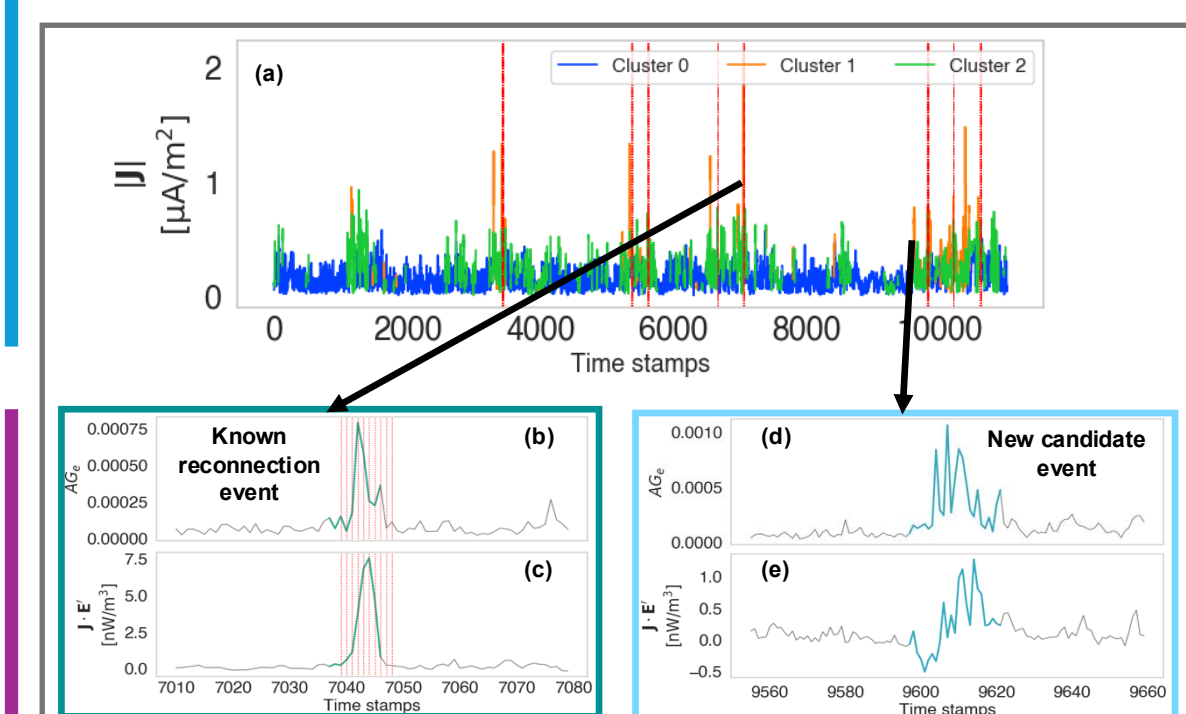
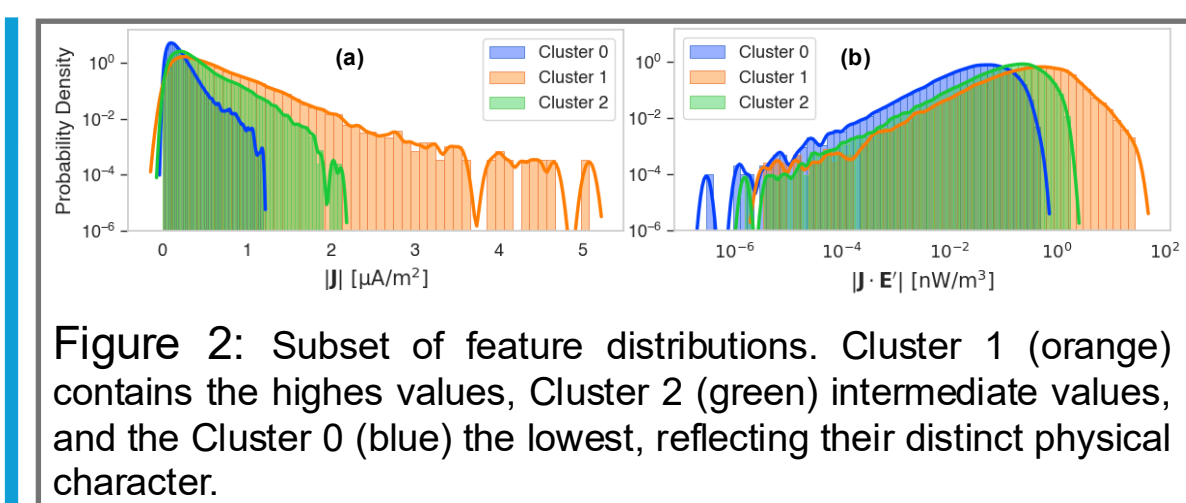


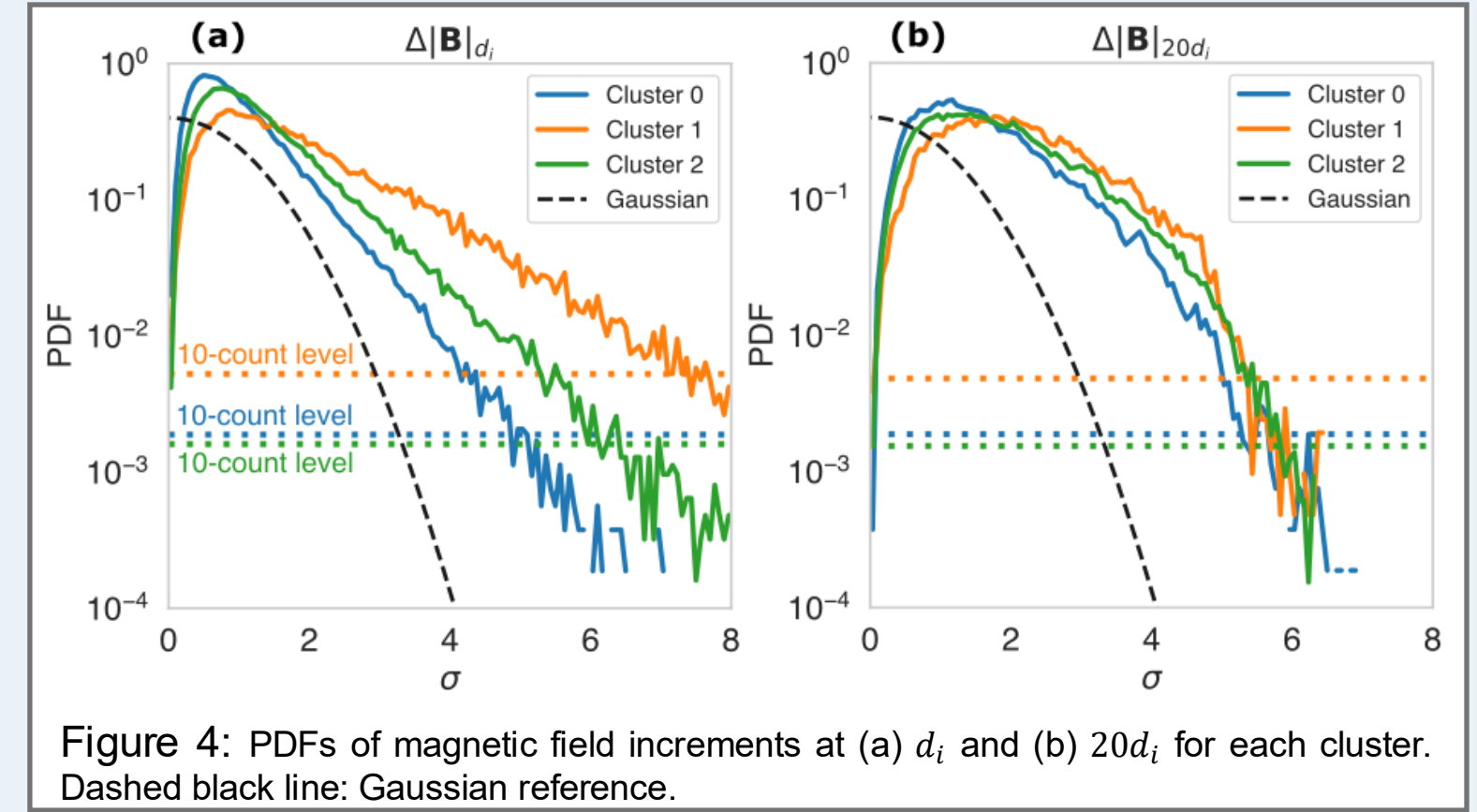
Figure 3: (a) Clustering results in the current density for an example turbulent interval. Reference reconnection events are shown in red lines. Zoom-ins of electron agyrotropy and energy conversion for (b-c) a known reconnection event and (d-e) a new reconnection candidate.

4 RESULTS

After characterising the TICC clusters, we examine whether they reflect distinct turbulent properties.

A. SMALL-SCALE INTERMITTENCY

- Increments measure fluctuations at scale ℓ : $\Delta Q = Q(x + \ell) - Q(x)$.
- At kinetic scales, $\ell = d_i$ (one ion inertial length), Cluster 1 shows the heaviest PDF tails (**Fig. 4a**), linking reconnection-like structures to extreme intermittent fluctuations.
- At larger scales, cluster distributions become more similar (**Fig. 4b**), suggesting that the strongest cluster separation occurs near kinetic scales.



→ Intermittency disappears at larger scales.
 → The ML algorithm isolates the most dissipative regions.

B. STRONGEST CROSS-SCALE ENERGY TRANSFER

The Local Energy Transfer (LET) estimates the local, scale-dependent rate at which turbulent energy is transferred across scales [Sorriso-Valvo+ (2018) *Solar Phys* 293]. Here, LET is computed at kinetic scales, ($\ell = d_i$), so large ($|LET|$) values indicate regions where the turbulent cascade is locally strongest

- The full $|LET_{1d_i}|$ PDF shows that Cluster 1 has the strongest tail, indicating enhanced local energy transfer (**Fig. 5a**).
- Peak $|LET_{1d_i}|$ value are computed within each identified structure to isolate the strongest events (**Fig. 5b**).
- Cluster 1 contains the largest LET peaks, linking reconnection-like structures to intense local cross-scale energy transfer.
- Cluster 2 shows intermediate LET values, while Cluster 0 remains closer to the Gaussian reference. However, both clusters exhibit similar peak-LET distributions.
- LET is not used in TICC, providing an independent physical diagnostic of the clusters.

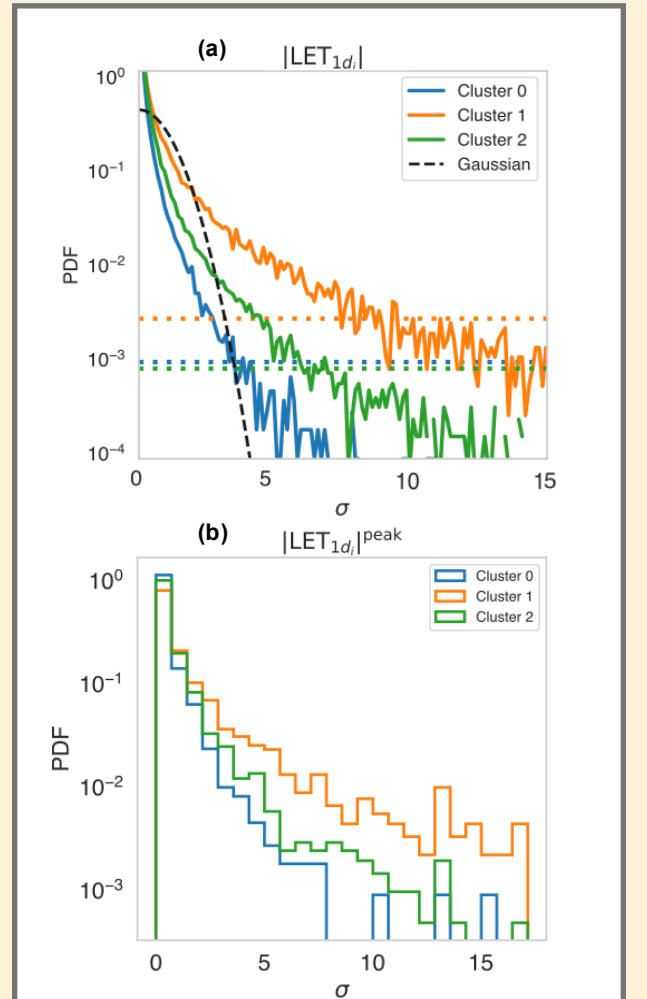
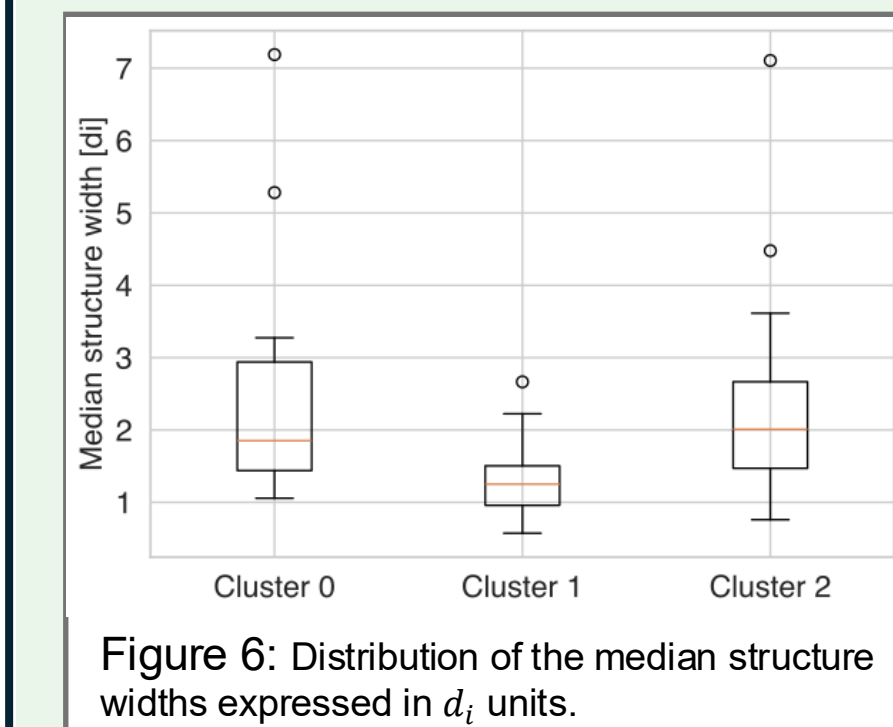


Figure 5: Distributions of (a) full $|LET|$ and (b) peak $|LET|$ values at one d_i for the three clusters.

C. LOCALISATION OF INTERMITTENT STRUCTURES



The most intermittent fluctuations are concentrated in localised structures with small characteristic widths, rather than being uniformly distributed throughout the turbulent interval (**Fig. 6**).

- Cluster 1 captures the sharpest, most compact structures.
- Cluster 2 captures broader complexes of intense structures that may also contribute to turbulent dissipation.

Combined with the increment and LET results, this suggests that TICC separates distinct coherent structures with different roles in the turbulent dynamics.

4 NEXT STEPS

Ongoing Work: Classifying reconnection candidates

Candidate structure from Cluster 1 are rotated into a local current-sheet-aligned coordinate system (LMN) using Hybrid Minimum Variance Analysis [Eriksson+ (2022) *ApJ* 933], making reconnection signatures become more distinct and can be compared with idealised reconnection profiles (red curves):

- B_L reversal (hyperbolic tangent fit) – **Fig. 7a**,
- Ion and electron outflow jets in $v_{i,L}$ and $v_{e,L}$ (Gaussian fit) – **Fig. 7b & c**,
- Magnetic Flux Transport Velocity [Li+ (2021) *ApJL* 909] (Gaussian and its first derivative fits):

$$U_{\Psi} = \frac{E_M}{B_{LN}} (\hat{M} \times \hat{b}_{LN})$$

→ U_L – **Fig. 7d**,

→ U_N – **Fig. 7e**.

The fitting parameters (e.g., jet amplitude, width of the jet, goodness of the fitting) encode physical information and serve as input for a secondary classification: false positives, standard and electron-only reconnection.

Preliminary results from 8 intervals:

311 candidates identified by TICC, with 164 new potential reconnection events.

Future Work

- Apply the method to a larger set of MMS magnetosheath intervals → build a large-scale reconnection event catalogue.
- Perform statistical analysis correlating reconnection occurrence rate with turbulence properties (e.g., correlation length, energy dissipation rate).

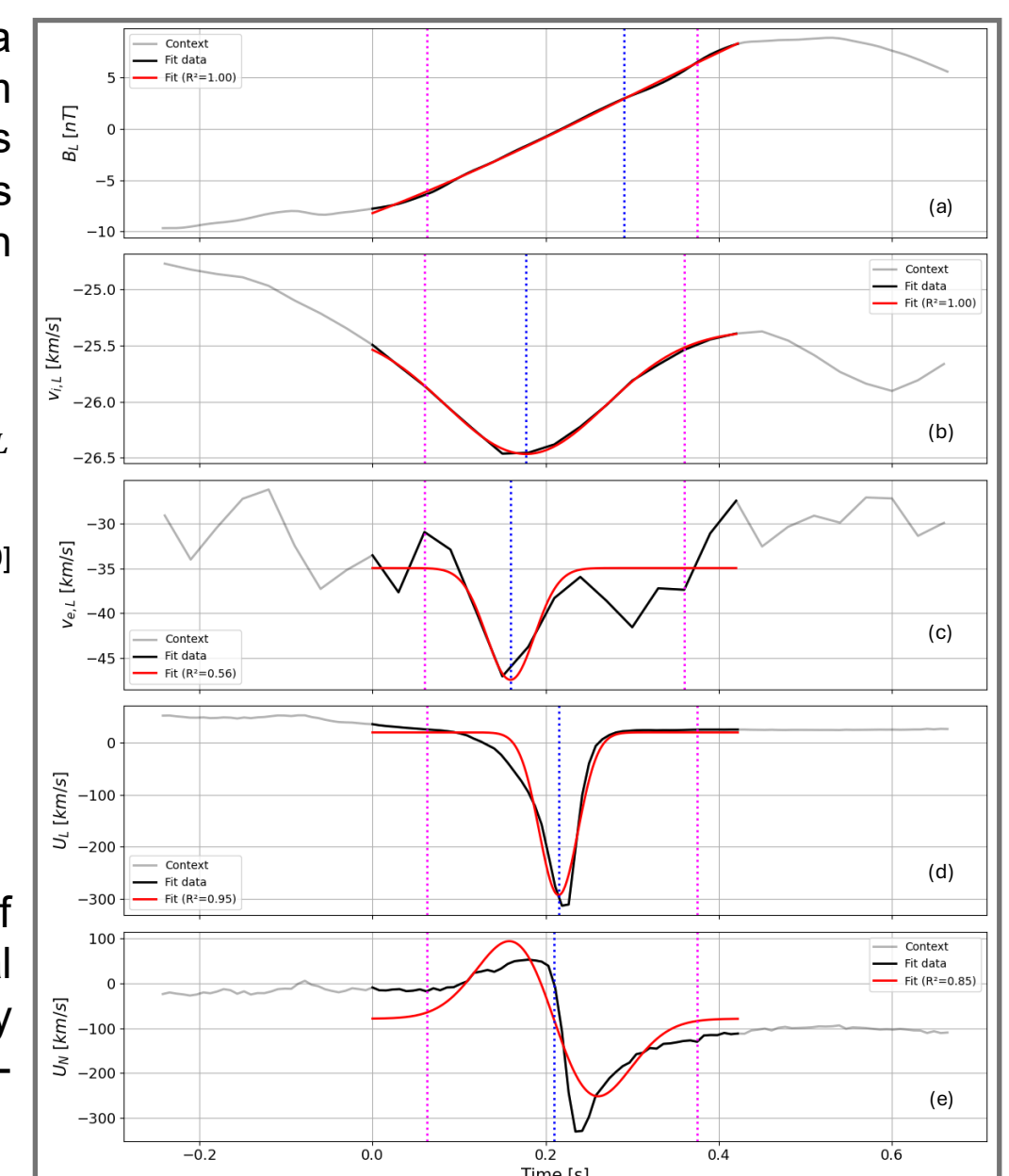


Figure 7: Example candidate event in the LMN coordinate system. Black curves show reconnection signature within the fitting window.